

Final Structure v3.0 — comprehensive

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Version genealogy and audit method

Five documents preceded this one, and v2.1 carried most but not all of them; the audit below checked every unit, principle and recommendation in each against v2.1 and found 38 items missing, weakened or unresolved, none of which required changing the four-level shape, so v3.0 is v2.1 plus those items and a set of structural rules.

Version	What it is	What it contributed that survives
v0 — the original syllabus	Six tracks, 39 modules, about 250 units, postgraduate register, Gulf-centred	The content at full depth; the Concept DNA method; the twelve grammars; the four boundary tests and blind interpretation; the studios' unit lists; the venture and release realism
The analysis (Instructional Analysis & Build Plan)	Diagnosis of v0; per-module essential questions; 25 interactive patterns; teaching frameworks; resource needs; gaps; build phases	Threads (Planetary Weight, Handover Grammar); the pattern library; threshold concepts; unit anatomy; the five criteria; the gap list; the pipeline phases
Concept v1	Rebuild for two audiences, global scope, self-paced, no budget	The two lanes; the artifact growing per unit; Level 0; the research plan by module type; the free-resource strategy; the AI partner's first three voices; the Module Factory

Version	What it is	What it contributed that survives
Concept v2	Full depth restored through three depths per module	Depth dials; vocabulary policy; nine angles; named routes; seed worlds; commissioning variants; build order; alternatives considered
Master Structure v2.1	Unit register: 21 modules, 10 tools modules, 10 engines, about 430 units	The register itself; the artifact schema; tools evolution paths; compound engines; media and lecture studies; research log rules

Audit method. Each earlier document was read unit by unit and recommendation by recommendation; each item was marked present, partial or absent in v2.1; partial and absent items received a fix (a new unit, a renamed unit, a rule, a thread or an appendix). Fixes were then checked for duplication against each other and against the register. The result is §2. Numbering stays stable: a new unit takes the number of the unit it follows plus a letter (F1.3a), so every reference in earlier documents still resolves.

Gap audit

Thirty-eight items; nineteen become new or renamed units, the rest become rules, threads, assets or appendices, and every one is traced to the version it came from.

#	Element	From	State in v2.1	Fix in v3.0
1	Interdisciplinary global discourse: what film, performance, sound, gaming and digital culture teach design	v0 1.1	Absent	New unit F1.3a
2	Vernacular intelligence: oral histories, folktales, local taxonomies as sources	v0 1.1	Only the regional slot	New unit F1.20a, with consent method from M1.5

#	Element	From	State in v2.1	Fix in v3.0
3	Identity, symbolism, collective memory	v0 1.1	Partial in F4.7	F4.7 renamed Semiotics, symbolism and collective memory
4	Temporal intelligence from literature, film, sound, performance	v0 2.1	Partial in M1.11	New unit M1.11a World-building lessons from film, games and performance
5	Practice DNA: the rules of the chosen medium	v0 2.2	Only implied by the O units	New unit in every area: Sx.1a The practice DNA of this medium
6	Omnichannel commerce and spatial retail: checkout, online-to-offline, the flagship as media	v0 5.5	Absent (dropped in v1)	New units R1.10a, R1.10b, R1.10c; Venture Console carrier layer
7	Cost, value and margin architecture	v0 5.9	Absent (engine layer with no unit)	New unit R2.4a
8	Expansion strategy across markets	v0 5.9	Absent	New unit R2.4b
9	Lifecycle economics: total cost of ownership	v0 5.9	Absent	New unit R4.4a
10	Data-rich models: BIM and PLM	v0 4.11	Only mentioned in S1.31	New unit T4.2a
11	Licensing agreements	v0 6.2	Absent	R2.6 renamed Contracts, NDAs, licensing, transfer of rights
12	Exhibition, museum and cultural-district experience	analysis §11	Absent	New unit S1.8a
13	Large-scale events, expos, activations	analysis §11	Partial in R1.10	New unit R1.12a (Mastery case)

#	Element	From	State in v2.1	Fix in v3.0
14	Service design, hospitality, food and drink	analysis §11	Partial in S1.11	New unit S1.8b The service script behind the space
15	Landscape, desert ecology, water	analysis §11	Only the regional slot	New units F5.5a Land, water and climate as the site; S1.24a The outdoor room
16	Game design and interactive narrative	analysis §11	Absent	Covered by M1.11a; S6 pattern library references game interfaces
17	Scent and culinary design	analysis §11	S1.21 only	Case dossiers added to S1.21; no new unit
18	Jewellery and accessories	analysis §11	Absent	New unit S3.35a Accessories and jewellery: the object on the body
19	Accessibility as a horizontal	analysis §11	Scattered	Thread 4, §8, with a checkpoint in every studio brief
20	Documentation and photography of physical work	analysis §11	Partial in S4.7, T5.3	New unit T5.1a
21	Client and stakeholder negotiation	analysis §11	Partial in R3.6	New unit R3.6a, Client voice
22	Writing and argumentation	analysis §11	Partial in R5.3	Thread 5, §8: one written argument per level
23	Synthetic worlds as a medium	analysis §11	S4.27–29, T3	Noted as future area S7 in §11
24	Research plan by module type	v1 §7	Absent	Carried into §10

#	Element	From	State in v2.1	Fix in v3.0
25	Unit anatomy and closing tests per module	v1 §9, analysis §9	Absent	§5 and §6
26	Teaching frameworks and threshold concepts	analysis §9	Absent	§5
27	The five assessment criteria	analysis §9	Absent (no grades)	Kept as the partner's five lenses, §9
28	Vocabulary policy	v2 §4	Absent	§5
29	Nine angles and named routes	v2 §5	Absent	Routes in §5; angles in §11
30	Seed worlds for learners without a project	v2 §5	M2.1 only	Asset: three seed worlds, §10
31	Commissioning variant of every studio brief	v2 §5	Absent	Rule in §5; CMB field in every Sx studio brief
32	Pattern library, three long-form patterns, Studio Workspace	analysis §8, v2 §7	Engines only	Build inventory §11; Studio Workspace added
33	Peer overlay when there is no cohort	v1 §9	Assumed	Rule in §9: pooled anonymised outputs, partner stand-in
34	Program voice from approved samples	v1 §10	Absent	Pipeline §11
35	Learner personas	v1 rhythm	Pending	Next steps §12
36	Commissioned regional assets	analysis §10	Absent	Program-produced assets list, §10, regional ones deferred
37	Institutions, partners, reading canon, kits	analysis §10	Absent	Appendix in §11: canon for our grounding, partners for later, things-at-hand instead of kits

#	Element	From	State in v2.1	Fix in v3.0
38	Hours by depth and prerequisites	analysis §2, v2 §3	Absent from the register	§6

Logic and efficiency review

The four-level shape and the depth dials hold; what v2.1 lacked was the connective tissue — threads, closing tests, prerequisites, reuse marking and a rule for briefers inside the studios — and those are the structural changes adopted here.

Finding	Problem it caused	Adjustment
Cross-cutting concerns (planetary weight, handover, attribution, accessibility, writing, AI collaboration) appeared as scattered units	A learner meets them six times without progression	Six horizontal threads with named checkpoints per level (§8); the units stay where they are, the thread gives them order
Modules had no closing test	Nothing marks a module as finished for either lane	A closing test per module, both lanes (§6)
No prerequisites in the register	Sequencing was implied, not stated	A prerequisite column (§6) and four sequencing rules (§5)
Labs reused across modules were indistinguishable from new ones	Build effort could not be counted	Reuse marked in the build inventory (§11): about 110 distinct labs across 25 patterns, plus 10 engines and 3 long-form patterns
Studio briefs served makers only	Briefers had no Mastery path in the areas	Every studio brief has a Make variant (ANC) and a Commission variant (CMB); the same site pack, a different task
Each area's medium-specific rules were implicit	The original's Practice DNA idea was lost	A Practice DNA unit at the start of every area's Practice tier

Finding	Problem it caused	Adjustment
Peer features assumed a cohort	Self-paced learners would hit empty overlays	Pooled, anonymised outputs from all learners feed the overlays; the partner's Stranger voice stands in when the pool is thin (§9)
The commerce, margin and expansion content was gone	The venture track could not answer "where does the money come in"	Five units restored (gap items 6–9)
Tools carried no data-rich modelling	BIM and PLM, the bridge to real handover, were missing	T4.2a
Voices were named inconsistently (Historian, Fabricator, CFO appeared without definition)	The partner had no spec	Seven voices defined once (§9)
Hours lived in v2, not in the register	Slices could not be sized	Hours by depth per module (§6)
Assets that cannot be free (regional readers, colour study) had no home	They were silently dropped	Program-produced assets list with what is done now and what waits for the regional vertical (§10)

Not changed, deliberately. The number of levels; the 21 + 10 module set; the six areas; the artifact schema; the regional slot as the last unit of every module; the no-grades rule; global first, region later.

Benchmarks

The closest institutional analogue is Central Saint Martins' MA Narrative Environments; the closest delivery analogue is Brilliant's lab-first, tutor-guided interactive model; v3.0 borrows structure from the first and mechanics from the second, and takes one thing each from four others.

Benchmark	What it does	What v3.0 adopts
CSM MA Narrative Environments	<cite index="2-1">Starts by charting contemporary and historical narrative environments, then critically demythologising, deconstructing, decolonising and decommodifying dominant narratives; systems observation and mapping is developed through counterfactuals and speculative histories to ground future propositions</cite>. <cite index="4-1">Its modules run from methodology and scope, through challenging, originating and repositioning, to a major project proposal, the major project and a critical report</cite>, and it <cite index="2-1">teaches budgets, project management, IP, collaboration, negotiation and proposal composition</cite>. It also maintains a networked course compendium of terms	The Foundations-to-Method order (chart, deconstruct, map, propose); a critical report as the written thread's capstone; the venture content as part of the design education, not an add-on; the glossary as a networked compendium rather than a list
Studio master's programs generally (RCA, DAE, the Bartlett)	Program specification and module descriptors with aims, outcomes, assessment and hours; external examination; the major project as the spine	Module descriptors (§6), the artifact as the major project, the defence as the examination

Benchmark	What it does	What v3.0 adopts
Brilliant	<cite index="13-1">Does not teach how to do something before asking questions; it lets learners try to find a solution before learning the procedure, and every lesson has practice sets</cite>. <cite index="11-1">A digital tutor sees the lesson and the problem a learner is working on and guides the reasoning step by step instead of revealing the answer</cite>; <cite index="11-1">courses are organised into learning paths with a suggested sequence</cite>	Lab before concept in every unit; the partner that questions rather than answers; routes as learning paths; practice sets as the module's closing test
World Building Institute and game-design pedagogy	The world as the deliverable; bibles that other departments build from; rules that generate play	The Concept DNA as a bible; the Stranger test as the proof that others can build from it
Cohort schools (IDEO U, Hyper Island-type)	Live critique and peer pods as the product	Deferred: optional paid crit cohorts once there are learners; peer pods as a later feature
AI tutoring products (Khanmigo-type)	Socratic tutoring inside the content, never grading	The partner's rules: questions, tests, no verdicts

What no benchmark offers. A single method spanning space, object, body, art, identity and digital product; two lanes on one spine; full studio depth in a self-paced format; a global-first world set with a regional vertical. That combination is the program's position.

The twelve principles of v3.0

Everything in the program follows from twelve rules; if a future change breaks one of them, the change is wrong or the rule needs a versioned revision.

1. **One curriculum, three depths.** Every module is written once at the original's depth. Orientation is its first fifteen minutes, Practice its usable core, Mastery its studio and defence. The Introduction, Practitioner and Comprehensive versions are slices by depth, never separate texts.
2. **Two lanes, one spine.** Makers produce; briefers write and judge. Same concepts, cases and labs; different Apply step and artifact attachments. Anyone may take any depth of any module. Every studio brief has a Make and a Commission variant.
3. **Four named routes.** Maker, Briefer, Founder, Student: published default paths through the depths, changing nothing structurally. A learner without a project takes one of three seed worlds.
4. **The artifact is the through-line.** The Concept DNA (schema in Master Structure §2) grows at the end of every unit, is tested at the end of every module, and is defended at the end of the program. Makers add anchor, family, adjacent translation and dossier; briefers add brief, judged set and commissioning record.
5. **The unit's shape follows its knowledge.** There is no fixed sequence. Each unit opens with the move that is most impactful for its particular knowledge and research (the opening moves below), and orders its parts to suit. Four things are invariant: every unit contains a lab or simulation, an Apply step by lane, a partner step, and a log entry; and a term is named after the learner has met it, wherever that is possible. Every kept term enters the glossary, which is a networked compendium.
6. **Tests, not grades.** Every module closes with a test (§6); the partner works through five lenses (generativity, craft, feasibility, cultural authority, evidence) and never issues a verdict.
7. **Seven voices.** The partner's voices (§9) replace the studio's critic, fabricator, client, finance director, historian, planet and stranger. They ask, test and reflect; they do not grade.
8. **Six threads.** Planetary Weight, Handover Grammar, Attribution and Authority, Accessibility, Writing and Argument, AI Collaboration run across all levels with checkpoints (§8), so recurring concerns progress instead of repeating.
9. **Precise vocabulary, no adjectives.** The profession's terms stay (typology, tectonics, affordance, palimpsest, tech-pack, MVI); manifesto intensifiers go; obscure headings are renamed (the v2 lists). The original's voice lives in provocations and studio briefs only.
10. **Free and open, with a record.** Every asset is sourced from open collections, public-domain texts, open data or free software, or produced by the program; every asset carries licence, source and the units that use it.

11. Global spine, regional slot. Every module's last unit is a regional slot; the world sets are global across the whole timeline; the Gulf, MENA and SWANA vertical fills the slots when it is built.

12. Evolving by version. Research is uncapped and logged; findings change the structure through recorded versions (Master Structure §12); new units take letter suffixes so references never break.

Opening moves (chosen per unit by the module designer, recorded in the unit spec)

Move	Opens with	Best when the knowledge is	Examples
Lab first	The interactive, no explanation	Perceptual, or a definition the learner can discover	F3.2 colour interaction; F2.1 archetype DNA; M2.2 the four tests
Object first	One object or case, told as a story	Historical, cultural, material	F1 centres; F4.1 material as text; S2.12 emotional durability
Story first	A scrollytelling narrative	Sequenced history, lineage, a transformation over time	F1.1 nine pivots; S3.11 silhouette and power; S1.14 palimpsest
Problem first	A brief, a constraint or a failure to fix	Procedural, technical, economic	S1.27 egress; S2.34 value engineering; R3.6 MVI
Prediction first	A question the learner answers before anything is shown	Warnings, counter-intuitive facts, ethics	F1.2 the spectrum; F5.8 what design cannot fix; T5.1 what a render claims
Argument first	The claim and its counter-claim	Theory that the learner already half-holds	S1.9 phenomenology; S4.12 relational theory; S2.10 critical design
Artifact first	The learner's own DNA or project, with a gap to fill	Method units late in a module; any Mastery unit	M2.17 full grammar authoring; M4.9 the anchor; R5.7 transfer brief

Move	Opens with	Best when the knowledge is	Examples
Partner first	A voice asks the question of the learner's work	Judgement and revision	M1.8 the insight; M4.7 abandoned directions; R5.3 defence
Data first	A dataset or a map to explore	Systems, climate, markets, routes	F5.3 feedback; F1.11 trade routes; R2.4b expansion
Making first	Hands on paper, card or the body before any screen	Tools, form, the body	T1.1 drawing; S2.19 the buck; S3.3 ease

The research log of each module says which move each unit takes and why; when a unit's research shows a different move would land harder, the unit changes, and the change is versioned.

Sequencing rules. Foundations before or alongside Method; Method before any area's Practice tier; an area's Mastery studio after M2 and M4 at Practice; tools attach to the unit that first needs them; venture modules after at least one area at Practice; R5 last. Experienced makers may open with M2 at Orientation as a preview.

Frameworks and threshold concepts (from the analysis, retained): backward design and constructive alignment; studio as signature pedagogy translated to the partner; threshold concepts taught slowly — history is a web, archetypes have DNA, the body is the metric, materials and words are signs, a concept is a generative system, generativity is testable, the collection precedes the object, the invisible systems are design material, circularity is the brief; cognitive apprenticeship (worked exemplar, then the learner's own); Kolb's cycle in the unit anatomy; mastery with re-submission instead of marks.

The program at module level

Twenty-one modules and ten tools modules; hours are Orientation / Practice / Mastery for a learner, including the additions in §7; unit counts are v3.0.

Foundations

Module	Units	Hours O/P/M	Prerequisites	Closing test — maker	Closing test — briefer	Engine
O1 How to see	4	2 / — / —	none	Read-a-Room annotation	same	—
F1 Histories of making	23	4 / 14 / 16	none	Object Autopsy on an unseen object, with attribution	Sort a case set on the reference-to- appropriation spectrum with reasons	Timeline Atlas
F2 Form	15	4 / 10 / 12	none	Dismantle an archetype, keep its identity	Judge five subversions: which keep identity, which break it	Typology Dismantler
F3 Body and senses	13	4 / 10 / 10	F2	Sensory rules for a scene in the Sense Engine	Audit a supplied scene against the body and colour rules	Sense Engine
F4 Material and meaning	14	4 / 10 / 10	F2	Material and verbal grammar candidates with reasons	Judge a set of material and copy choices against a stated meaning	—
F5 Systems and consequences	12	3 / 8 / 10	F4	Ecosystem map and one scenario	Same, plus what the intervention cannot fix	Planetary Ledger

Method

Module	Units	Hours O/P/M	Prerequisites	Closing test — maker	Closing test — briefer	Engine
M1 Research and premise	17	3 / 11 / 12	Foundations at O	A premise defended against the Client voice	Same, on a premise written as a brief	—
M2 Concept DNA	20	4 / 14 / 16	M1	The Stranger test: drift scored, one rule fixed, rerun	The Stranger builds from the brief; the briefer judges with the four tests	Stranger Lab

Module	Units	Hours O/P/M	Prerequisites	Closing test — maker	Closing test — briefer	Engine
M3 World architecture	10	3 / 10 / 12	M2	Cutting Room: subtract a third of the collection with reasons	Judge a supplied collection's roles and coherence	Collection Matrix
M4 Make and test	12	2 / 10 / 30	M3	Adjacent translation with drift under threshold	Commission the partner as designer; judge three directions; revise the brief once	area engine

Practice areas

Area	Units	Hours O/P/M	Prerequisites	Closing test — maker	Closing test — briefer	Engine
S1 Space and experience	38	2 / 17 / 72	M2 (M4 for Mastery)	Anchor on the site pack; handover audited	Commission variant of the brief; judged set of five spaces	World Engine
S2 Object and product	41	2 / 16 / 60	same	Object family from buck to BOM	Commission variant; judged set of five objects	Object Line
S3 Body and wearables	48	2 / 16 / 72	same	Capsule with tech-pack	Commission variant; judged set of five garments	Capsule
S4 Art and installation	38	2 / 16 / 50	same	Site proposal with documentation plan	Commission variant; judged set of five works	Site Work
S5 Identity and communication	25	2 / 16 / 50	same	Identity system a stranger can apply	Commission variant; judged set of five identities	Identity Engine
S6 Digital product and interface	21	2 / 16 / 50	same	Prototype with tokens and accessibility audit	Commission variant; judged set of five flows	Product Loop

Tools

Module	Units	Hours P/M	Attached to	Closing test
T1 Seeing and drawing	5	8 / 12	F2, S1.7	An orthographic set read by the Handover Auditor
T2 Parametric thinking	5	6 / 16	F2.13, M3	One rule regenerating a coherent family
T3 Working with AI	5	8 / 12	M2.15	A brief that reproduces the DNA under blind interpretation
T4 Simulation, interface, data-rich models	6	8 / 18	F3.8, S1.31	A walkthrough or prototype of the studio project
T5 Honest evidence and documentation	5	6 / 12	M4, S4.7	An evidence set with a representation-ethics statement
T6 Reality capture	3	— / 14	S1.5, S2.19	A clean capture used in an engine
T7 Data visualisation	4	— / 14	M1, F5	An interactive view of the learner's data
T8 Physical computing	4	— / 14	S2.30, S3.25	A sensing and acting behaviour, simulated or built
T9 Environmental analytics	2	— / 14	F5.6, S1.19	A climate simulation and an LCA of the studio project
T10 Fabrication	3	— / 12	S1.17, S2.20	A part with its toolpath and failure analysis

Venture and release

Module	Units	Hours O/P/M	Prerequisites	Closing test — maker	Closing test — briefer	Engine
R1 The world as venture	16	2 / 11 / 12	one area at P	Orbital architecture and carrier strategy defended	Judge a brand architecture for drift from its DNA	Venture Console
R2 Money, law and rights	13	2 / 11 / 12	R1	Two funding scenarios run; IP route chosen with reasons	Same, from the commissioner's side	Venture Console
R3 Making it real	13	2 / 10 / 12	one area at M for Mastery	Handover package audited; MVI negotiated with constants intact	The brief as artifact, audited; negotiation from the client side	Handover Auditor
R4 Responsibility	9	2 / 8 / 8	R2	Final planetary ledger with data	Judge a release plan's lifecycle claims	Planetary Ledger
R5 Defence and dossier	9	2 / 8 / 20	everything	Transfer brief executed; defence in the Defence Room; dossier	Transfer brief as commission; defence of the judged sets; dossier	Defence Room

Totals, v3.0. About 450 units. Introduction slice about 48 h; Practitioner slice with one area about 215 h; Comprehensive with one studio and two tool electives about 500 h.

Unit additions and changes register, v3.0

Twenty-two units are added and two renamed; the Master Structure remains the single unit register (so two copies cannot drift), and these rows are appended there under a v3.0 entry. Columns match the register.

Unit	Concept	Depth	Field	Delivery	Interactive · media	Follows
F1.3a	Interdisciplinary discourse: what film, performance, sound, gaming and digital culture teach design	P	SRC, GRM.n	L, C	Cross-Discipline Case Wall · vid, aud, img	F1.3

Unit	Concept	Depth	Field	Delivery	Interactive · media	Follows
F1.20a	Vernacular intelligence: oral history, folktale and local taxonomy as sources, with consent	M	SRC	L, K, D	Oral History Kit (consent, prompts, transcription) · aud	F1.20
F4.7 (renamed)	Semiotics, symbolism and collective memory	P	GRM.n	L, C	Sign-Swap · img	—
F5.5a	Land, water and climate as the site	P	PER, PLW	L, C, I	Site Systems Map · dat	F5.5
M1.11a	World-building lessons from film, games and performance: bibles, rules, stakes, time	P	GRM.n, GRM.t	L, C	Bible Anatomy (dissect a public world bible) · vid	M1.11
S1.1a	The practice DNA of space: what the medium adds and forbids	P	ALN	L, I	Practice DNA card	S1.1
S1.8a	Exhibition, museum and cultural-district experience: object, story, visitor	P	GRM.s, GRM.n	C, I	Visitor Journey Mapper · img, drw	S1.8
S1.8b	The service script behind the space: hospitality, food and drink, operations	P	GRM.b, HND	C, I	Service Blueprint Builder	S1.8a
S1.24a	The outdoor room: landscape, shade, water, ground as spatial practice	M	GRM.s, GRM.a	C, S	World Engine, outdoor layer · dat	S1.24
S2.1a	The practice DNA of the object	P	ALN	L, I	Practice DNA card	S2.1
S3.1a	The practice DNA of the garment	P	ALN	L, I	Practice DNA card	S3.1
S3.35a	Accessories and jewellery: the object on the body	M	GRM.f, GRM.k	C, K, I	Fit Tester, body mode · img (open collections)	S3.35
S4.1a	The practice DNA of the installation	P	ALN	L, I	Practice DNA card	S4.1
S5.1a	The practice DNA of identity	P	ALN	L, I	Practice DNA card	S5.1
S6.1a	The practice DNA of the interface	P	ALN	L, I	Practice DNA card	S6.1

Unit	Concept	Depth	Field	Delivery	Interactive · media	Follows
T4.2a	Data-rich models: BIM and PLM, cost and tolerance inside the geometry	M	HND	L, I	Data-Rich Model Explorer · 3d	T4.2
T5.1a	Documenting physical work: photography, capture, sequencing	P	DOS	L, K, I	Documentation Planner · img	T5.1
R1.10a	The physics of the checkout: navigation, friction, payment	P	VEN	C, I	Checkout Flow Dissector	R1.10
R1.10b	Online-to-offline synchronisation	P	VEN	I	O2O Journey Mapper	R1.10a
R1.10c	The flagship as media	P	VEN	C, I	Media-Value Calculator · img	R1.10b
R1.12a	Large-scale activation: expos, festivals, civic events	M	VEN	C, S	Activation Workflow, scale mode · img	R1.12
R2.4a	Cost, value and margin architecture: from cost of goods to price	P	RGT	L, S	Venture Console, margin layer	R2.4
R2.4b	Expansion across markets: regulation, behaviour, logistics	P	RGT	L, I	Expansion Canvas	R2.4a
R2.6 (renamed)	Contracts, NDAs, licensing, transfer of rights	P	RGT	I	Contract Clause Explorer	—
R3.6a	Client and stakeholder negotiation	P	HND, CMB	S, P	Negotiation Simulator; Client voice	R3.6
R4.4a	Lifecycle economics: total cost of ownership	P	LCY, RGT	S	Venture Console, lifecycle layer	R4.4

Rules changes that touch existing units. Every Sx studio brief unit gains a Commission variant (CMB). T4 is retitled Simulation, interface and data-rich models; T5 is retitled Honest evidence and documentation. S1.21 gains scent and culinary case dossiers. F1.18 and R2.8 are the same protocol at two depths and say so.

The six threads

A thread is a concern that recurs across levels and must progress rather than repeat; each has a named checkpoint per level and one owner engine or tool.

Thread	Level 1	Level 2	Level 3	Tools	Level 4	Owner
1. Planetary Weight	F5.6 rough ledger; F5.10 first ledger	M3.6 cost and risk distribution includes ecological weight	Sx circularity units; the engine's carbon layer	T9.2 full LCA	R4.6–R4.7 final ledger; R4.4a lifecycle economics	Planetary Ledger
2. Handover Grammar	F2 orthographic language via T1	M4.4 construction logic; M4.8 specialist gaps	S1.32, S2.34, S3.40, S6.18 area packs	T4.2a data-rich models	R3.8, R3.11 the audited package	Handover Auditor
3. Attribution and Authority	F1.2 spectrum; F1.16 labour; F1.18 protocol; F1.20a vernacular sources	M1.5 ethics and consent; M1.14 fieldwork	S2.11 craft translation; S3.15, S3.46 regional dress; S4.13 mythologies	T3.5 bias and sovereignty	R2.7–R2.8 protocol and responses	Historian voice
4. Accessibility	F3.3 the body as metric; F3.4 cognitive load	M2.13 category obligations include access	S1.27 egress; S3.30 adaptive; S5.16 contrast; S6.6, S6.16 audits; a checkpoint in every studio brief	T5 evidence readable by all	R3.9 compliance	Accessibility Checker
5. Writing and Argument	One argument card answered per Foundations module	M1.8 the insight; M1.16 the defended premise	Each studio brief's statement of intent	T5.1a documentation	R3.3 pitch; R5.3 critical defence; the critical report in the dossier	Defence Room

Thread	Level 1	Level 2	Level 3	Tools	Level 4	Owner
6. AI Collaboration	F2.13 parametric first rule	M2.15, M2.18–19 Stranger and brief architecture	Sx.21 or equivalent: guidelines a stranger can apply; engines' generative layers	T3 all	R3.2 studio OS; R5.7 transfer brief	Stranger Lab

How a thread is used. The app shows each thread as a progress line across the learner's route; a module cannot close while its thread checkpoint is empty; the dossier lists the six threads as six short chapters.

The AI partner

Seven voices, one rule: they ask, test and reflect, and never say good or bad; between them they stand in for the critic, the fabricator, the client, the finance director, the historian, the planet and the stranger a studio would otherwise need in the room.

Voice	Asks	Where it appears	What it can see
The Mirror	"Here is what your work is doing; is that what you meant?"	Every Apply step; M1.8; M4.7	The learner's artifact and the unit's concept
The Stranger	"I have never met you; here is what I built from your rules."	M2.15, M2.18; every Sx.21-type unit; R5.7	The DNA or brief only, never the conversation
The Client	"I am commissioning this; what do I get, what does it cost, what could go wrong?"	Briefs; M1.16; R3.3; R3.6a; S4.8	The DNA, the brief, the venture fields
The Fabricator	"This will not build; here is why, and what it would take."	M4.8; every studio brief; R3.6; T10	The engine state, materials, tolerances
The CFO	"This costs that; cut this and you break these constants."	S1.29, S2.34, S3.32; R2; R3.12	The Venture Console and the DNA constants

Voice	Asks	Where it appears	What it can see
The Historian	"Where does this come from, who made it, whom did you ask?"	F1; M1.14; S2.11; S3.15; R2.7	The world set, the sources, the attribution record
The Planet	"What does this weigh, and where does it go when it dies?"	F5; Sx circularity; R4	The Planetary Ledger and the BOM

Rules. No verdicts; every response ends with a question or a test the learner can run. The five lenses (generativity, craft, feasibility, cultural authority, evidence) shape what each voice looks at, not a score. Voices declare their agenda. Everything generated is logged, attributable and scoreable for drift. Free-tier limits are shown, not hidden.

Standing in for peers. Overlays (the cohort's sorts, thresholds, ratings) draw on a pooled, anonymised set of all learners' outputs; when the pool is thin, the Stranger produces the comparison set and says so. Peer exchange in M2.18 is the Stranger until real exchange is available; live crit cohorts are a later, optional product.

Standing in for specialists. The Fabricator, CFO and Historian carry the specialist consultations the original studios assumed; each has a knowledge pack per area (materials and processes, cost norms, source lists) that the research logs feed.

Research, resources, media and lecture rules

Research is uncapped and typed; sources are free and recorded; media is mostly sourced and sometimes made; lecture material is whatever must be told, kept to a fifth of the time and always preceded by a lab.

Research plan by module type

Type	Modules	What it needs	How it is done	What you check
Historical and cultural	F1, F4, M1.11a, Sx lineage units, S5.11, S6.9	World sets of 40–60 objects and cases with open images, dates, provenance, makers; scholarship; public-domain primary sources	Scans of open-access scholarship through the academic connectors; object selection from CC0 collection APIs; object stories written from the record and cited	Balance across regions and eras; fairness; five facts per module spot-checked
Perceptual and skills	F2, F3, T1–T5	Stable principles; good examples; the labs	Principles cross-checked once; examples from open collections or the learner's own; labs built and tested	Whether the lab produces the perception it claims; wording for non-designers
Method	M1–M4	Exemplar grammars and collection logic; the program's worked world	Public brand and design-system guidelines, game bibles and pattern languages read for structure; the exemplar DNA written and tested in the Stranger Lab	Taste and coherence of the exemplar

Type	Modules	What it needs	How it is done	What you check
Practice	S1-S6, T6-T10	Global case sets; the medium's constraints; free-tool documentation; free standards	Case scans; open 3D and drawings; tutorials written against the tool's own docs; practitioner voices from public talks with attribution	Realism: what a practitioner would object to
Venture	R1-R5	Plain-language public guidance on money, IP, commerce, handover; open templates	WIPO and national IP guides; public finance and commerce guides; open contract and brief templates; every unit labelled literacy	That nothing reads as advice; formulas visible

Research outputs per module (always the same four files): source list with licences; case library entries (object, image link, licence, date, place, maker, story, units); claims sheet; open questions for you. Research logs and sprint entries live in the Master Structure §12.

Program-produced assets (what the analysis called commissioned; now made by the program from open sources, in this order)

Asset	Serves	Now or later
The world set: about 1,200 objects and cases, tagged by unit, with a regional slot per module	all	Now, module by module
The exemplar Concept DNA and three seed worlds	M1-M4, S1-S6	Now, before M2

Asset	Serves	Now or later
The networked glossary (compendium), with "say it to a designer" lines	all	Now, grows per module
Argument cards (claim, counter-claim, case) for every theory unit	F, M, Sx theory units	Now, per module
Site, object, body and brand packs per area	Sx studio briefs	With each area
Partner knowledge packs per area (materials, processes, cost norms, sources)	the Fabricator, CFO, Historian	With each area
A global history reader in the program's own voice	F1	Now, as F1 is built
Regional readers: history, materials, colour and light, dress	regional slots	Later, with the regional vertical

Free sources and rules (from Concept v1, unchanged): CC0 museum collections and their APIs for images; public-domain texts for primary sources; open-access scholarship; open data; CC 3D; Google Fonts; open-source and free-tier software with a named fallback for every free tier; embedding rather than re-hosting video; a licence note on every asset; paraphrase first, quotations short, no summaries that replace a copyrighted book; assets referenced by link where licence requires.

Media rules (from Master Structure §10): a unit's media list is part of its spec; nothing is Drafted without media sourced or scheduled; every asset records licence, source and units; production is limited to drawings, animated diagrams, engine recordings, synthesised audio and the packs.

Lecture rules (from Master Structure §11): six kinds of lecture material (sequenced narrative, theory and argument, definitions and taxonomies, procedures, cases, warnings); each has a conversion (navigate before reading, argument card then lab, sort then reveal, the tool is the procedure, cases as dossiers, ethics as simulators); text first, video only where motion carries meaning, nothing over eight minutes; about a fifth of learner time.

Reference canon (for our grounding, not for learners, since it is not free): the reading lists in the analysis §10 stand; the program's own copy replaces them for learners, with attribution.

Presentation forms for explanatory content

Anything that must be explained gets a form chosen for the kind of explanation it is, from a catalogue of eighteen; layered, animated and scroll-driven pages are the default for narrative and theory, step-through explainers for procedures and systems, and plain text is the fallback, not the norm.

#	Form	What it is	Best for (lecture kind)	Example units	Free stack	Effort
1	Scrollytelling page	Multi-layered, parallax page: images, drawings or 3D as depth layers, text pinned and revealed as the reader scrolls; strata, time and scale become literal layers	Sequenced narrative; theory that has a spatial or temporal shape	F1.1 nine pivots; F1.11 routes; S3.11 silhouette history; S1.10 courtyard to tower; F4.6 patina; S1.14 palimpsest	CSS scroll-driven animations; a scroll library; three.js for 3D layers	M-L
2	Step-through explainer	A diagram or scene that builds one step at a time; the learner advances, can go back, can branch	Procedures; systems; taxonomies	F5.3 feedback loops; M2.6 the twelve grammars; S1.27 egress; R3.8 handover anatomy; S2.26 tolerances	SVG plus a small state machine	S-M
3	Object story	An annotated image with hotspots, deep zoom and toggled layers (material, labour, trade, meaning)	Cases	Every C unit; the Object Autopsy's reading mode	IIIF deep-zoom viewers (OpenSeadragon) on museum images	S

#	Form	What it is	Best for (lecture kind)	Example units	Free stack	Effort
4	Explorable explanation	Prose with live variables the reader drags inside the sentence; the text recomputes	Definitions with thresholds; anything with a formula	M2.4 thresholds; F3.6 colour physics; R2.2 dilution; S1.29 yield	in-app components	S-M
5	Before/after and morph	A slider or scrub between two states, or across time	Change over time; subversion; reuse	F2.13; F4.6; S1.5; S3.18	image compare; morph targets	S
6	Animated diagram	A short looping motion graphic with no narration, next to the text	Concept cards; mechanisms	F2.10 grids; F3.5 joints; S2.18 kinetic states	SVG or Lottie-type animation	S
7	Short video	Under eight minutes; only where motion carries meaning; engine screen recordings; embedded public talks	Motion-dependent content; practitioner voices	S3.16 drape; S2.19 the buck; S4.20 light as mass	screen capture; embeds	S-M
8	Audio companion	Narrated track to listen while doing; field-walk guides; argument cards read aloud	Making units; walks; revision	M1.14 field sprint; S1.9 phenomenology walk; any argument card	synthesised narration	S
9	3D turntable and walkthrough embed	A model or site the reader orbits or walks, inside the page	Form, scale, space	F2.11 scale; S1.31; S2.16	three.js viewer	M
10	Comparison wall	Filterable grid of cases side by side, with a shared lens	Case sets; judging practice	Every module's judged set; S5.11; S6.9	in-app grid	S

#	Form	What it is	Best for (lecture kind)	Example units	Free stack	Effort
11	Timeline scrub	The Timeline Atlas in reading mode: drag through time, layers on and off	Narrative history	F1; S3.11; S5.11	Timeline Atlas	reuse
12	Interactive map	Places, routes and flows	Trade routes; sites; expansion	F1.11; S1.24; R2.4b	map library on OSM	M
13	Argument card	Claim, counter-claim, one case, expandable to the essay	Theory	S1.9; S4.12; F2.4; S2.10	in-app card	S
14	Sequential illustration	Comic-strip panels for rituals, use choreography, service scripts, audience journeys	Procedures with people in them	S2.4; S1.8b; S4.30; M1.11	drawn panels, in-app	M
15	Prediction prompt	A question the reader answers before the reveal, inline in the text	Warnings; definitions; any claim worth testing	throughout	Guess-and-Reveal pattern	reuse
16	Guided sketch-along	Draw with the page: overlays and steps on the learner's own canvas	Drawing and notation	T1; F2.1; S1.7	Drawing Trainer	reuse
17	The designed brief	A studio brief laid out as a document with the program's voice, printable	Studio briefs	every Sx studio brief	in-app, exportable	S

#	Form	What it is	Best for (lecture kind)	Example units	Free stack	Effort
18	Two-voice dialogue	Historian and Client (or Fabricator and CFO) argue the unit's question, generated and optional	Theory and warnings for listeners	argument units; R2; R3	partner voices, synthesised audio	S

Rules for the forms. Choose by lecture kind, not by taste: narrative → 1, 11, 3; theory → 13, 1, 18; definitions and taxonomies → 2, 4, 15; procedures → 2, 14, 16; cases → 3, 10; warnings → 15, 14, 18. Parallax and depth layers are used only where the layers mean something (strata, time, scale, inside and outside), never as decoration. Every animated form has a reduced-motion version and a text alternative; every page reads on a phone; nothing autoplays with sound. One form per lecture element, two at most; the unit's opening move (§5) decides whether the form or the lab comes first. Every form is a component built once and filled with content, so the catalogue is part of the build inventory (§11).

Consumption matrix

What a learner needs to learn comes in five kinds, and how they consume it comes in six situations; every unit is published in three consumption forms so that the same content meets the learner on a phone for five minutes, at a desk for an hour, and in the hand while making.

What they need to learn (by level and lane)

Kind of knowledge	What it is	Heaviest in	Maker needs	Briefer needs
Know-that	Facts, histories, names, dates, sources	Foundations, lineage units	Enough to place their own work	Enough to talk to designers and check claims
Know-why	Theory, argument, judgement	Foundations, Method, venture	To defend choices	To judge choices

Kind of knowledge	What it is	Heaviest in	Maker needs	Briefer needs
Know-how	Procedures, moves, tool skills	Method, areas, tools	The full procedure	The procedure's shape, to brief and audit it
Know-who and know-where	Sources, makers, communities, institutions, archives	Attribution thread, regional slots	To credit and consult	To commission and credit
Dispositions	Habits: test before trusting, subtract, log changes, ask who made it	Everywhere; the partner's questions	Both	Both

How and where they consume (situations, with what fits each)

Situation	Time	Attention	What fits	Forms	Depth
Phone, in transit	5–10 min	Fragmented, one hand	A provocation, one lab, one card, one prediction prompt	15, 13, 3, 6, 11 in scrub mode	Orientation
Desk, focused	45–90 min	Full	The whole unit: lab, concept, cases, apply, partner	1, 2, 4, 9, 10, the labs	Practice
Hands busy, making	30–120 min	Ears free, eyes on the work	Audio companion, step-through on a second screen, the designed brief printed	8, 2, 17, 16	Practice, Mastery
Walking or on site	20–60 min	Outdoors, moving	Audio walk, field prompts, capture tasks	8, 12, capture tools	Mastery (M1.14, S1.9, S4.26)
Reviewing before a decision	10–20 min	Skimming, comparing	Comparison wall, glossary compendium, the artifact's own fields, the judged set	10, 13, the DNA Builder read mode	Any
Revisiting as reference	2–5 min	Look-up	Glossary entry, card, checklist, formula panel	13, 4, checklists	Any

Three published forms per unit. The five-minute form (Orientation layer: the unit's opening move, one lab or card); the session form (the full unit anatomy); the reference form (glossary entries, cards, checklists and the artifact fields the unit filled). The app serves the form by device and session length; a learner can switch. Studio briefs also exist as printable documents; making units have an audio companion; field units have an offline pack.

Modes of consumption a unit should offer (at least three of six): read, watch, listen, play, make, browse. Foundations units lean read-play-browse; Method units play-make-read; studio units make-play-listen; venture units play-read-browse.

By career stage and lane (defaults, not limits): a student reads and plays most; a mid-career designer makes and skims theory; a founder browses, plays the simulators and listens; a briefer reads cards, browses walls and plays the judging labs. Routes (§5) set these defaults; the three published forms make them possible without duplicating content.

Design consequences. Every unit's spec now carries its consumption forms and its presentation forms alongside its media list; the glossary is built as reference-first (look-up in two taps); the Orientation layer is written to be complete on a phone; audio is generated for every argument card and every making unit; and the pattern library gains the presentation catalogue as a second component set.

Build inventory, pipeline, rhythm and alternatives

The program is built from 25 reusable patterns, 3 long-form patterns and 10 engines, instantiated as about 110 distinct labs; the pipeline runs through this account's tools with a human at the two points that need one, and the build order proves the spine before the studios.

Build inventory

Kind	Count	Examples	Priority
Reusable patterns (analysis §8)	25	Structured Editor (DNA Builder), Generative Lab, Spectrum Sort, Guess-and-Reveal, Calculator, Systems Map, Matrix Planner, Revision History, Parametric Morph, 3D Sandbox, Sequence Timeline, Role-Play, Synthesis Board, Card Deck, Decision Tree, Bilingual Text, Layered Timeline, Layer and Time, Scenario Generator, Journey Mapper, Routing Puzzle, Node Playground, Drawing Trainer, Simplified Physics, Unfolder	Eight priority-1 patterns ship the spine
Long-form patterns	3	Studio Workspace (project, site pack, prototypes, revision history, partner threads in one place); Handover Auditor; Defence Room	With the first studio
Compound engines (Master Structure §9)	10	Timeline Atlas, Sense Engine, Stranger Lab, World Engine, Object Line, Capsule, Site Work, Identity Engine, Product Loop, Venture Console	Stranger Lab, Sense Engine, Timeline Atlas first
Distinct labs	about 110	pattern instances with content; reuse marked in the register	Per module
Partner voices	7	with knowledge packs per area	Mirror, Stranger, Client first

Pipeline (from Concept v1, retained): concept and curriculum in Claude Docs; research through web, Exa, Firecrawl, Elicit, Scite, Consensus, Scholar Gateway, LlamaParse, Hugging Face; program voice through the brand-voice skills; visual design in Figma and the artifact Design type; labs as artifacts with the web-artifacts, frontend-design and algorithmic-art skills; the app on Lovable and Supabase (Base44 as fallback); learner testing with the user-research skills; assets in Drive or Dropbox; operations by scheduled tasks and the Module Factory skill; quality through accessibility, validation and code-review skills. Human points: taste and fairness; the last look at facts and rights.

Roles I play; what stays with you. I draft, research, build and revise; you decide, judge, spot-check, recruit the five testers and set the voice. Regional partners, guest critics and institutional relationships are listed in the analysis §10 for the day they are wanted.

Things at hand instead of kits. Paper, card, a phone camera and the learner's own body, kitchen and wardrobe cover Foundations, Method and the Practice tier of every area; Mastery studios list an optional kit (foam, callipers, a half-scale form, a microcontroller) with a no-kit route through the engines.

Working rhythm (2–3 days a week): a weekly digest of what moved; one module in the factory at a time; research sprints logged; structure versions recorded; your review day closes each module's stage.

Build order. Fix the DNA schema and pack format → Practice tier of Foundations and Method → Mastery of M2 and M4 → Practice tier of all six areas → Mastery studio for S2 → Practice tier of venture → Mastery of R3 and R5 → remaining studios and tool electives through the factory → the regional vertical.

Alternatives considered (Concept v2 §6, retained for the record): two products; core plus pathways; studio-first; region-first. The chosen shape is depth dials with named routes. **Nine angles** for reading the program (Concept v2 §5) remain a checklist for every future version: lane, depth, artifact, medium, knowledge type, traditional versus innovative, geography, career stage, time budget. **Future area S7, synthetic worlds as a medium**, is noted for when S5 and S6 exist.

Change log and next steps

v3.0 is the working baseline: this document holds the framework, the Master Structure holds the unit register with the v3.0 additions appended, and every later change is a versioned entry in both.

Change log

- v3.0 — Audit of five versions; 38 items restored or resolved; 22 units added, 2 renamed; six threads, closing tests, prerequisites, seven partner voices, commissioning variants, reuse marking; hours by depth restated; benchmarks recorded. Unit count about 450.
- v3.1 — Added the presentation-forms catalogue (18 forms with rules) and the consumption matrix (five kinds of knowledge, six situations, three published forms per unit); unit specs now carry consumption and presentation forms; the build inventory gains the presentation catalogue as a component set.
- v3.2 — Principle 5 revised: no fixed unit sequence; each unit opens with the move that suits its knowledge (ten opening moves catalogued), with four invariants (a lab or simulation, an Apply by lane, a partner step, a log). Presentation-form and consumption rules updated to match.

What is still open

1. Learner personas (one maker, one briefer, one founder, one student) in their own words.
2. The program's name and the artifact's name.
3. The exemplar world: invented, or something of yours.
4. Confirmation of F1 as the first module through the factory, M2 second, S2 the first studio.
5. The six decisions from Concept v2 §8 that remain provisional: depth dials with named routes (assumed yes); six areas (assumed yes, original four built first); Method before Foundations allowed (assumed yes); first studio S2 (assumed yes); the vocabulary lists (assumed agreed).

Pass 2 opens when you confirm item 4. Its first output is F1's research log, not a lesson.